

Class Test 1
CER3C4-Digital Electronics
Computer Engineering sec A and sec B

Time: 70 Min

Max Marks: 20

Note: All Questions are compulsory.

- Q.1 **Explain why is a two-input NAND gate called universal gate?** 4
- Q.2 **Convert the following** 4
- (a) $(1011.10)_2$ to $(\dots\dots\dots)_{10}$
 - (b) $(524.23)_{10}$ to $(\dots\dots\dots)_8$
 - (c) $(236.75)_8$ to $(\dots\dots\dots)_2$
 - (d) $(A69.2B)_{16}$ to $(\dots\dots\dots)_{10}$
- Q.3 **Find out** 5
- (a) **9's complement of $(155)_{10}$ and 10's complements of $(155)_{10}$.**
 - (b) **2's complement of $(1010.11)_2$ and 1's complement of $(1010.11)_2$.**
- Q.4 **Explain AND and OR logic in details. Perform subtraction $(M-N)$ using r's complement of the given problem.** 4+3
- (a) $M=534$ and $N=2451$
 - (b) $M=1011$ and $N=110$